Nikhil Mehta

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RESEARCH INTERESTS

Machine Learning, Deep Learning, Optimization, Robotics

EDUCATION

Delhi Technological University, New Delhi, India — 2010-2014
B.Tech. in Software Engineering with **First Class**
Percentage: 74.8

St. Francis De Sales, India
All India Senior School Certificate Examination, Class XII, 85% — 2010
All India Secondary School Examination, Class X, 89% — 2008

MANUSCRIPTS UNDER REVIEW

Nikhil Mehta, James R. McBride & Gaurav Pandey (2017), **"Robust and Fast Scan Alignment for Autonomous Vehicles using Mutual Information"**, International Conference on Intelligent Robots and Systems (IROS), Vancouver Canada.

RESEARCH EXPERIENCE

Learning based Localization of Autonomous Vehicles — Ongoing
Guided by Prof. G. Pandey at IIT Kanpur
Currently working on a discriminative learning based approach for localization of an autonomous vehicle. Image registration of ground reflectivity and z-variance maps is done using a convolutional neural network implemented using Tensor Flow.

MI based Scan Registration for Autonomous Vehicles — Oct - Feb '17
Guided by Prof. G. Pandey at IIT Kanpur
Implemented a fast Mutual Information based scan alignment algorithm on cuda (GPU implementation) for autonomous vehicles. Speed and robustness was found to be better than current state of the art scan matching methods on two real time dynamic datasets recorded using a Lidar. Also worked on a super-voxel clustering algorithm for 3D point clouds to represent an efficient Gaussian Mixture Model (GMM).

INDUSTRY EXPERIENCE

Software Engineer at Fitso — Jan - Aug '16
Worked on a real time tracking and video application, allowing consumers to improve their general fitness. Was among the first 5 in the tech team. Our work here had an impact on more than 50,000 daily users seeking to improve their fitness. Application was given the Top Developer badge by Google after six months of launch.

Software Engineer at Zomato — June '14 - Jan '16
Developed server side APIs and Android application for various consumer and enterprise products provided at Zomato across 22 countries including India, US and Canada. Developed a classification algorithm for restaurant reviews for spam filtering using neural network.

PROJECTS & INTERNSHIPS

Research Intern at University of Birmingham, UK — Dec '13 - Jan '14
As a research intern at the Human Computer Interaction Center, I developed a paradigm and an experimental game in Python to investigate how people's perception of a hidden partner affect their behavior while competing against an artificial intelligent bot. Audio instructions using speech recognition and text summarization was also integrated using Python NLTK.

Research Intern at University of Leeds, UK Jul - Aug '13
As a research intern at the School of Computing, designed a vision algorithm for cell (lamallepodia) tracking in Differential interference contrast (DIC) microscopic videos using entropy differences in subsequent frames. Real-time implementation was further improved by implementing Kalman filter.

Severity prediction using data mining and machine learning
Course project guided by Prof. Ruchika Malhotra at DTU Jan - May '14
Extracted multiple features from various open source dataset documentations using tf-idf, bag-of-words and word2vec. Applied multinomial regression and logistic regression for severity classification and prediction.

Hierarchical Reinforcement Learning for Line Follower
Robotics club at DTU Mar - Dec '13
As a part of the robotics club at DTU, worked on the movement-planning layer for a line follower that had to find the optimal path to reach the goal position in a complex maze. Our implementation allowed policy learning directly from the experience by decomposing the problem into a hierarchical sub-structure.

3D Modeling and stitching of scenes using ICP
Senior year project guided by Prof. Ruchika Malhotra at DTU Jan - May '14
Used an extension of iterative closest point for registration and remeshing to create a 3D model of environment using a Kinect sensor.

Mobile based university attendance
Junior year project guided by Prof. Ruchika Malhotra at DTU Aug - Sep '12
Made an android application that allowed instructors to use their smartphone to mark attendance, publish announcements and set reminders for classes. All the data was synced to the web, providing a easy integration between the web and mobile application. This project was selected in the "Role of Technology in Saving the Environment" conference at DTU.

TECHNICAL SKILLS

Languages: C, C++, Java, CUDA for GPU, Python, Tensor Flow for Deep Learning, MATLAB, LaTeX.
Web Development: HTML5, CSS, JavaScript, PHP, SQL.
Software: Git {github-handle: nikhil-dce}, Eclipse
Operating Systems: Linux, OSX, Android.
Proficient Algorithmic Skills

RELEVANT COURSEWORK

Computer Science: Data Structures, Algorithms, Artificial Intelligence, Machine Learning, Parallel GPU Programming using CUDA, Compilers
Mathematics: Convex Optimization, Linear Algebra, Probabilistic Models

POSITION OF RESPONSIBILITY

- Was appointed as the **Technical Head** in the Society of Software Engineering in Delhi Technological University (DTU)
- Member of organizing council of the Software Engineering fest in DTU